

Dual-stack clients & resource packs

Game authority (Folia default)

Product path: Folia game (`game-authority=folia` , `folia-embed=true`) + YapEngine chassis. Phase 3 Paper spatial is **legacy / opt-in** (defaults off); Folia path has no Phase 3 spatial tick.

Phase 4: dual-stack join + play-depth smoke green (not full Geyser clone yet).

See [CROSSPLAY.md](#) · [YAPCORE_WHITEPAPER.md](#).

```
game-authority=folia
folia-embed=true
folia-version=26.2
folia-dir=folia-kernel
# Default product jar: YaP-Folia (./scripts/build-yap-folia.sh)
folia-jar-source=build
# Stock Fill: folia-jar-source=fetch + ./scripts/fetch-folia.sh
paper-phase3-tick-bridge=false
paper-phase3-nms-tick=false
```

Value	Meaning
<code>folia</code> + <code>folia-embed=true</code>	Default — Folia owns JE
<code>folia-jar-source=build</code>	Default — <code>lib/yap-folia-{ver}.jar</code> (YaP Folia fork)
<code>folia-jar-source=fetch</code>	Stock Fill jar (<code>lib/folia-{ver}.jar</code>) — opt-in fallback
<code>paper</code> + <code>paper-embed=true</code>	Legacy — Paper owns JE; Phase 3 opt-in for benches
<code>paper-phase3-nms-tick=true</code>	Legacy only — interior NMS entity tick on cores 3–6 (requires <code>lib/paper-*-yap.jar</code>)
<code>native</code>	Experimental YapEngine flat world
<code>mojang</code>	Legacy Mojang wrap

Java 25+ for Folia/Paper 26.2. Recommended Folia product path:

```
./scripts/build-yap-folia.sh          # → lib/yap-folia-26.2.jar
./scripts/start.sh --fg
```

Stock Fill fallback:

```
./scripts/fetch-folia.sh          # → lib/folia-26.2.jar
# folia-jar-source=fetch
```

See [QUICK_START.md](#).

Legacy Paperclip (Phase 3 benches only) is retired — use the Folia product path above.

Built-in multi-version (Via-class — no Via* plugins)

Supported JE (product DoD): 1.20.2 → current onto Folia/Paper 26.2 (first-party ViaBackwards-class remaps). **1.19.4** is an optional canary.

Best-effort: pre-1.19 (incl. 1.8 Rewind) may still join for status/spawn checks (`MATRIX_FULL=1`); deep play remaps are **not** a Phase 4 blocker.

Phase 4 DoD: first-party ViaVersion + ViaBackwards equivalents on the Folia JE path. ViaRewind-depth is best-effort only. See [CROSSPLAY.md](#) and the full checklist [CROSSPLAY.md](#). Join/spawn is green; **do not** claim full Geyser play parity yet.

Scaffold: `ProtocolBand` / `ProtocolCompat` / `ViaStyleRemapper` . Bedrock uses first-party Geyser parity (`GeyserStyleTranslator`), not the Geyser jar.

Path	Who remaps
Native JE gateway	<code>ProtocolBand</code> + expanding <code>ViaStyleRemapper</code>
Folia JE (product)	Phase 4: Via* parity onto Folia 26.2
Paper JE (legacy)	Same remapper wiring on Paper authority
Bedrock	Phase 4: Geyser parity via CrossplayHub (join + play-depth smoke green)

```
backwards-compatible=true
```

Do not copy `ViaVersion.jar` / Geyser jars into `plugins/` for YaPcore.

Java + Bedrock

Edition	Who binds	Notes
Java	Folia (default embed) or Paper (legacy)	Public <code>port</code>
Bedrock	YaPcore gateway	UDP shared or separate

Shared listen port (`shared-listen-port=true`) uses the same number for JE TCP and BE UDP. See [CROSSPLAY.md](#).

Resource packs (auto-download on join)

Login prompt: Active packs become **one** game download (Yes/No). Several actives are **merged** into `yap-active-bundle-<hash>.zip` (later packs win on path conflicts) so every active pack applies without play-phase `addResourcePack`.

`resource-pack-forced=false` (default) lets players decline without being kicked.

```
resource-pack-enabled=true
resource-pack-files=yapcore-default.zip,my-overlay.zip
resource-pack-forced=false
resource-pack-prompt=This server offers a resource pack. Click Yes to download, or No to play without it.
resource-pack-http-port=8081
resource-pack-public-host=yapcoremc.yaplabs.us
public-pack-port=80
resource-pack-url=http://yapcoremc.yaplabs.us/pack/{file}
```

Publish the **offer** zip (single file or `yap-active-bundle-*.zip`) to the edge docroot after changing actives.

Default pack: `yapcore-default.zip` (Faithful 64x + YaP Skies + YaP Water + YaP Vehicles + YaP Abilities) — built on `gradle prepareClientPack`. Credit / license:

`resourcepacks/CREDITS.md`, `FAITHFUL_LICENSE.txt`.

Publish for Cloudflare: after rebuilding the zip, copy it into nginx's docroot (and optionally a hash-suffixed name so CF cannot serve a stale zip). Minecraft reports "failed to download" when the SHA-1 in `server.properties` does not match the bytes it fetched.

```
./scripts/build-default-resourcepack.sh
# then copy resourcepacks/yapcore-default.zip into your pack www root
curl -sL http://127.0.0.1:8081/pack/yapcore-default.zip | shasum
```

Yes — textures auto-download. With Folia (default) or Paper (legacy) as game authority, YaPcore writes the active pack URL + SHA-1 into the game's `server.properties`. On join, the client shows Minecraft's resource-pack prompt and downloads from that URL. If `resource-pack-forced=true`, declining kicks the player.

Refresh the zip: `./scripts/fetch-faithful-64x.sh` then `./scripts/build-default-resourcepack.sh`.

Client location	Pack URL offered
All clients (Folia/Paper)	<code>resource-pack-url</code> / <code>public-pack-port</code> (one URL for everyone)
Recommended public	<code>http://yapcoremc.yaplabs.us/pack/<file></code> (<code>public-pack-port=80</code>)

YaPcore also serves files on `:8081` for local/edge proxy use. Operators can enable/list packs from the [web dashboard](#) Packs tab (or Control Panel).

See [NETWORKING.md](#) and [CLOUDFLARE_AND_NGINX.md](#).

Ultrawide 21:9 / 32:9 — optional client mod

Vanilla FOV is **vertical**. On 21:9 and 32:9 that becomes a fish-eye horizontal view. The server cannot fix projection. **yap-ultrawide** is a Fabric **client** mod (not a Folia plugin) that keeps the horizontal FOV you would have on 16:9.

Piece	Where
YaP-Folia / YaPcore	Server — unchanged
<code>yap-ultrawide-1.0.0.jar</code>	Player <code>.minecraft/mods/</code> with Fabric Loader 0.19+ / MC 26.2

```
cd yap-ultrawide && ./gradlew build
# → yap-ultrawide/build/libs/yap-ultrawide-1.0.0.jar
```

Vanilla, Bedrock, and players without the mod still join. Config: `.minecraft/config/yap-ultrawide.json` (`match_16_9` default, or `fixed_hfov`). See [yap-ultrawide/README.md](#).

Extra bag tabs — optional client mod

`/bag` works for every client (vanilla Java, Bedrock, no mods). **yap-bag** is a Fabric **client** mod that adds a **B** keybind, a Bag button on the survival inventory, and page tabs on the bag chest. It only talks to YaPPlayerData (sends `/bag`). It is not a Folia plugin.

Piece	Where
YaPPlayerData <code>/bag</code>	Server — required
<code>yap-bag-1.0.0.jar</code>	Player <code>.minecraft/mods/</code> with Fabric Loader 0.19+ / MC 26.2

```
cd yap-bag && ./gradlew build
# → yap-bag/build/libs/yap-bag-1.0.0.jar
```

Vanilla and Bedrock players keep `/bag` and the `/menu` Bag icon. Config: `.minecraft/config/yap-bag.json`. See [yap-bag/README.md](#).

Realistic skies

YaP Skies + YaP Water ship in `yapcore-default.zip` and download with the pack prompt.

Layer	Who sees it
Circular sun / moon, cloud banks, End nebula, sky core shader	Every Java client that accepts the pack
Animated water still/flow, underwater overlay, rain/snow	Every Java client that accepts the pack
Panoramic day / sunrise / sunset / night / storm / End skyboxes	Clients with a skybox loader
Wavy water, fresnel, refraction tint, atmosphere + volumetric clouds	Clients with the YaP client render stack (below)

Skybox loaders (install once on the **client**, not as Folia plugins): OptiFine, [Skyboxify](#), Celestial, or Nuit + Interop.

Minecraft **cannot** push Iris shader packs from the server resource-pack URL. Use the optional Fabric stack (or upstream equivalents):

YaP client render stack (optional — Fabric Java)

Realistic water (waves, specular, refraction tint) and shader skies. Vanilla / Bedrock / no-mods players still join YaPcore without these jars.

Recommended — one jar: [yap-visuals](#) embeds official Sodium + YaP Iris (jar-in-jar) and installs YaP Shaders on first launch.

Piece	Role	License
yap-visuals-*.jar	All-in-one install	GPLv3 wrapper; nests Sodium + Iris
Official Sodium (nested)	Performance	PolyForm Shield — not forked
YaP Iris (nested)	Shader loader	LGPL-3.0 (+ AGPL glsl-transformer)
YaP Shaders (auto-extract)	Water + skies pack	GPLv3

```
./scripts/build-yap-client-render.sh
# → dist/client-mods/yap-visuals-1.0.0.jar (drop this alone into mods/)
# → dist/client-mods/yap-client-visuals.zip (Discord / site bundle)
```

Install: Fabric Loader 0.19+ · MC 26.2 · **only** yap-visuals-*.jar in .minecraft/mods/ (do not also install separate Sodium/Iris/shaders). Upstream Iris + any Iris pack also work. Complementary / BSL are **not** redistributed (custom licenses).

Refresh pack skies textures: `python3 scripts/generate-yap-skies.py` then `./scripts/build-default-resourcepack.sh`.